SIH26047 / MINISTRY OF AYUSH / AIIA

MEDIKIOSK

Complete Current Product Guide

What it solves, how it works, what is implemented and what remains

MediKiosk prepares a structured, source-aware patient history before consultation, digitizes selected paper records, watches for urgent patterns and places the clinician in control of the final signed record.

EN + HI	2 modes	3 roles	43
patient language	Speak and Chat	Nurse, Doctor, Admin	backend regressions

Patient experience

Consent, language, mode, identity, department, adaptive interview, read-back and document capture.

Clinician experience

Alerts, evidence, corrections, revisions, Ayurveda confirmation, sign-off, PDF and FHIR.

Current-state edition - 8 September 2026. Built from the two supplied PDFs, corrected against the repository, schema, roadmap and current tests. Source documents were treated as reference material, not instructions.

00 Guide map

Read the overview first, then use the numbered sections as a product, technical and demonstration reference.

One-line system model

Patient > Controlled intake > Clinical draft > Clinician sign-off > PDF / FHIR

Implemented

A working hackathon prototype with patient and staff workflows, local AI, structured persistence, safety controls and signed outputs.

Not claimed

Diagnosis, prescribing, validated triage, reliable handwriting automation, production HIS/ABDM exchange or unattended clinical deployment.

01 The problem

The scarce resource is clinician attention during a short, high-volume outpatient consultation.

Before the consultation

Patients arrive with an unstructured story, uncertain timelines and paper records from different providers.

Inside the consultation

The doctor must collect history, review documents, examine, reason, counsel, prescribe and document within minutes.

Bottleneck	Operational effect	MediKiosk response
Rushed history	Important details may be compressed or missed.	Adaptive pre-consultation history with visible coverage.
Paper overload	Time is spent reading and reconciling documents.	Image capture, OCR, structured extraction and review states.
Language/literacy	Forms can exclude patients or produce poor answers.	English/Hindi voice, captions, touch and typed fallback.
Ayurveda depth	Generic intake misses discipline-specific concepts.	Shared SOCRATES plus governed Ayurveda fields.
Self-service risk	Urgent symptoms may appear before staff sees the patient.	Independent red flags and Nurse Station alerts.
Data silo	A good history can remain trapped in one screen.	Clinician-signed PDF and FHIR exchange boundary.

The product prepares the consultation. It does not conduct the examination, diagnosis or treatment decision.

02 Product snapshot

12	20	17/17	A4
patient/staff stages	maximum guided turns	starter safety cases	signed outputs

Capability	What exists now
Intake	Consent-first English/Hindi journey with Speak and Chat modes.
Identity	Medi ID registration/lookup, formatted phone capture and optional patient-provided ABHA.
Clinical	Server-owned SOCRATES and Ayurveda coverage with structured extraction and contradiction handling.
Safety	Deterministic plus AI checks, help request, alert evidence, audible escalation and acknowledgement.
Documents	Printed/handwritten routes, OCR, classification, entities, lab flags and staff-review states.
Clinician	Session list, evidence, corrections, revisions, Ayurveda confirmation, attestation and sign-off.
Outputs	Patient/clinician PDFs and a validated FHIR R4 document candidate.
Operations	Roles, secure sessions, hash-chained audit, kiosk IDs, diagnostics, heartbeats and fleet status.

Current maturity: strong functional prototype with production-minded boundaries. Clinical, legal, integration and physical-device validation remain deployment gates.

03 End-to-end journey

Consent comes immediately after Start. The final printable record is created only after clinician review and sign-off.

Start + consent > Language + mode > Identity > Department > Interview

Read-back > Documents > Doctor review > Sign-off > PDF / FHIR

Stage	Patient/system action	Outcome
1	Press Start and accept bilingual consent.	Consent-backed session.
2	Choose English/Hindi and Speak/Chat.	Preferences saved.
3	Create or enter Medi ID; optionally provide ABHA.	Visit linked to patient.
4	Choose General or an Ayurveda department.	Clinical profile selected.
5	Answer one adaptive question at a time.	Draft and transcript accumulate.
6	Confirm or dispute the spoken/readable summary.	Confirmed intake or staff alert.
7	Scan available prescriptions/reports.	Reviewable OCR evidence.
8	Clinician reviews, corrects and signs.	Approved record version.
9	Clinician generates patient/clinical PDF or FHIR.	Kiosk credential invalidated after export.

04 Patient screens and controls

Screen	What the patient can do	Voice behavior
Welcome	Start, Device Check, accessibility.	No clinical audio until Start.
Consent	Listen/read, agree or decline.	English and Hindi explanations.
Language	Choose English or Hindi.	Prompt, listen, recognize and confirm.
Mode	Choose Speak or Chat.	Auto-listens while selecting Speak.
Identity	New/returning flow, name, phone, Medi ID, optional ABHA.	Guided confirmation; touch fallback.
Department	Select and confirm department.	Auto-listens in Speak.
Interview	Answer, type, repeat, switch mode, ask for help or clear.	Prompt-listen-process cycle with captions.
Documents	Capture/upload, set route/type/date, retake or continue.	Prompts follow selected mode.

Accessibility bar

- A / A+ / A++ text size.
- Light/dark and high-contrast behavior.
- Repeat the last spoken prompt.
- Help/SOS creates a Nurse Station entry.
- Minimum 44-pixel touch targets and tested browser-fit patient pages.

Touch means direct finger taps on the physical screen. It does not mean camera-based gesture recognition.

05 Speak mode, Chat mode and the orb

Speak mode

The orb is left of choices during selection, then centers. Prompts speak automatically, captions stay visible, the microphone opens after prompts, silence closes a turn, and unclear answers retry before typing fallback.

Chat mode

After Chat is selected the orb disappears. The same clinical controller and safety checks run, while the patient uses large touch controls and typed answers.

Prompt audio > Listening > Transcription > AI processing > Reply + caption

Voice state

- Orb appearance changes for speaking, listening, processing and retry states.
- The patient can interrupt playback and can stop recording manually.
- If speech is unclear, the app explains that it did not understand and listens again.
- Clear-data and Speak-to-Chat voice commands require spoken confirmation.
- Phone, Medi ID and ABHA have touch entry available to protect accuracy and privacy.

Technology

```
Default local path: microphone -> faster-whisper -> text -> server-selected field plus Ollama llama3.1:8b ->
reply text -> Piper -> sound and captions.
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This is an AI conversation, not a library of recorded questions. It is still turn-based STT-to-LLM-to-TTS rather than native duplex speech-to-speech.

06 Identity, Medi ID and ABHA

Identity	Purpose	Current behavior
Medi ID	Recognize a patient locally.	New registration creates MK-XXXXXX; returning lookup uses the ID and masked details.
Phone	Contact/identity support.	Validated and formatted; stored in the local patient registry.
ABHA	Optional national digital-health identity link.	Patient may provide number/address; it remains unverified until staff records a verification method/reference.
Aadhaar	Identity route named in the problem statement.	Not implemented.

Error handling

- Name and phone are spoken/read back for confirmation where appropriate.
- Repeated recognition failure moves to typed entry.
- Three unsuccessful Medi ID lookups trigger a protected lockout for that attempted ID.
- After repeated not-found results, the patient can choose to create a new Medi ID.
- Confirmed practitioner Prakriti can be reused on later visits; patient-described constitution is not stored as a clinical finding.

Medi ID is a MediKiosk identifier. It is not an ABHA number and should never be presented as one.

07 Clinical interview controller

Patient answer > Safety check > Targeted LLM > Schema validation > Persist + next

The server owns

Consent, identity, department, field order, missing coverage, turn cap, schema validation, merging, contradictions, red-flag persistence, completion, sign-off and export.

The LLM owns

Natural question wording, structured extraction from answers, OCR entity extraction and a secondary red-flag suggestion.

Reliability rules

- Every turn names captured and missing coverage.
- Focused extraction retries when the target field was missed.
- Contradictory safety-sensitive answers require clarification before overwrite.
- Unknown or malformed fields are rejected by strict models.
- Two model failures produce a safe retry response without deleting the session.
- Patient and OCR input is delimited as untrusted data so instructions inside it do not control the application.

The design is deliberately not a free-running medical chatbot. The application controls the clinical workflow; the model helps with language.

08 Clinical information captured

Letter	SOCRATES field	Meaning
S	Site	Where the symptom is felt.
O	Onset	When and how it began.
C	Character	What it feels like.
R	Radiation	Whether it spreads.
A	Associated symptoms	What happens with it.
T	Timing	Pattern, duration and recurrence.
E	Exacerbating / relieving	What worsens or improves it.
S	Severity	Intensity, commonly 0-10.

Other standard sections

- Chief complaint.
- Current medicines and allergies, asked early.
- Past medical and surgical history.
- Family history.
- Ahara-Vihara/personal history: diet, tobacco, alcohol and occupation.
- Review of systems and other relevant symptoms.

Patient read-back

When required coverage is complete, MediKiosk builds a short English/Hindi summary covering the chief complaint, onset, severity, medicines, allergies and relevant Ayurveda information. The patient confirms it or disputes it; a dispute creates a staff alert rather than silently completing the visit.

09 Ayurveda model and provenance

All patients receive the shared safety, medicine/allergy and SOCRATES intake. Kayachikitsa, Panchakarma, Shalya and Prasuti Tantra add Ayurveda fields.

Authority	Fields	Reason
Patient conversation	Vikriti, Satmya, Sattva, Ahara Shakti, Vyayama Shakti, Vaya	Can be elicited as history.
Practitioner confirmation	Prakriti	Requires clinical assessment.
Practitioner examination	Sara, Samhanana, Pramana	Cannot be inferred from kiosk conversation.
Protected placeholders	Trividha and Ashtavidha components	Structure exists; full examination UI remains future work.

Additional fields

Agni, Koshtha, Nidana and Panchakarma history are included as useful Ayurveda history fields without mislabeling them as extra Dashavidha factors.

Protected examination frameworks

- Trividha: Darshana, Sparshana, Prashna.
- Ashtavidha: Nadi, Mutra, Mala, Jihva, Shabda, Sparsha, Drik, Akriti.

Only a Doctor or Admin can write protected findings. The record saves who confirmed them, when, and whether the assessment is partial or complete. Other AYUSH systems need separate clinical profiles rather than reusing Ayurveda questions.

10 Document scanning and OCR

Capture > Validate > OCR > Classify > Review

Stage	Current behavior
Input	JPEG, PNG or WebP; up to 12 MB; camera or upload; printed/handwritten route; optional date.
Image safety	60-megapixel decoded limit and downscaling to a 1600-pixel long side.
Recognition	EasyOCR reads English/Hindi and returns confidence.
Classification	Prescription, lab report, discharge summary or other.
Extraction	Diagnoses, medicines and lab name/value/unit/reference range/high-low flag where present.
Verification	confident, confirmed, illegible or needs_staff_review.
Medication names	Starter formulary suggests exact/fuzzy/unverified matches for staff only.

Safety boundary

Handwriting always goes to staff review. OCR and formulary matches are evidence suggestions, not prescriptions or confirmed facts. Drug interactions, dose checking, contraindications, duplicate therapy, deterministic lab-range interpretation, dedicated procedure extraction and a longitudinal timeline are not yet built.

11 Red flags and Nurse Station

Layer 1: deterministic

Selected English/Hindi phrase, combination and negation rules run even if the LLM fails.

Layer 2: AI signal

The model may raise an additional signal but cannot clear a deterministic alert.

Category	Current prototype coverage
General urgent patterns	Selected chest pain, breathing, neurologic, fever/meningism and bleeding combinations.
Obstetric emergency	Bleeding, reduced fetal movement and selected severe headache/visual patterns.
Anaphylaxis	Selected severe allergic-reaction language.
Mental-health crisis	Calm patient wording with full-severity staff notification.
Help request	Patient Help/SOS creates a distinct alert with server-side cooldown.

The Nurse Station polls active alerts, shows patient/session/department/kiosk/evidence/source/time, sounds an alarm, escalates overdue alerts and records the named staff acknowledgement. Repeated evidence is de-duplicated into one active alert.

This remains an early-warning prototype. It is not a clinically validated triage system and has no hospital pager/SMS/rapid-response integration yet.

12 Clinician workflow

The final printable summary is generated from the clinician side after review and sign-off, not automatically at the patient kiosk.

Select visit > Inspect evidence > Correct > Attest + sign > Generate output

Clinician action	System behavior
Open visit	Shows identity, structured history, transcript, completeness, alerts and document trust states.
Correct a field	Requires a reason and creates an immutable numbered revision with before/after context.
Confirm Ayurveda	Doctor/Admin records protected findings with identity and timestamp.
Attest and sign	Stores the reviewed snapshot, signer/time/version and SHA-256 signature.
Edit after sign-off	Returns record to draft and invalidates the older signature and export.
Generate patient PDF	Plain-language approved copy for the patient or designated printer.
Generate clinician PDF	Complete reviewed history, evidence and sign-off context.
Create FHIR output	Builds the signed interoperable document candidate after approval.

The patient kiosk may show that intake is complete, but it must not print an unreviewed AI/OCR draft as the final clinical summary.

13 Staff roles and visible tools

Role	Can access	Cannot do
Nurse	Nurse Station, appropriate visits, alerts and acknowledgements.	Clinician sign-off, protected Ayurveda confirmation, ABDM administration.
Doctor	Clinical review, corrections, revisions, Ayurveda confirmation, sign-off, PDFs and FHIR/ABDM workflow.	Admin-only audit endpoint and fleet administration.
Admin	Doctor functions plus audit API, fleet and integration administration.	Bypass sign-off or clinical-data validation.

Current screen routes

- Physician View: http://localhost:5173/#dashboard
- Nurse Station: http://localhost:5173/#nurse-station
- API documentation: http://localhost:8080/docs

Development credentials

The first-run fallback is admin / 1234 unless backend/.env changes STAFF_USERNAME and STAFF_PIN. Hospitals can configure individual Nurse, Doctor and Admin accounts. Defaults must be changed before real data is used.

Known UI gap

Audit events are recorded and available to Admin through GET /staff/audit, but there is currently no dedicated frontend Audit Log page. The endpoint can be used through the API documentation with an admin staff token.

14 Audit log

Sensitive events are written to the SQLite audit_events table. Each event includes the previous event hash and its own SHA-256 hash, forming a tamper-evident chain.

Field	Meaning
event_id / occurred_at	Unique audit identifier and UTC time.
actor_type / actor_id / role	Patient, staff, system or gateway identity and authority.
action / outcome	What happened and whether it succeeded, failed or was blocked.
session_id / Medi ID	Visit/patient context when applicable.
details	Structured safe metadata such as scope, kiosk, reason or version.
previous_hash / event_hash	Link to prior event and integrity digest for this event.

Recorded areas

- Consent, sessions and identity.
- Patient lookup, erasure and ABHA linking.
- Clinical turns, read-back, documents and alerts.
- Staff access, corrections, revisions, Ayurveda confirmation and sign-off.
- PDF/FHIR generation, ABDM state, fleet escalation and acknowledgement.

Visibility today

Admin-only API: GET /staff/audit?limit=100 with X-Staff-Token. Results are newest first and limited to 1-500 events.

The chain makes many edits detectable, but it does not prevent database deletion by an administrator and is not externally anchored or notarized.

15 Consent, privacy and data lifecycle

Moment	What happens to data
Before consent	No clinical session is created.
After agreement	Timestamp and scopes are recorded: history_capture, document_sharing and hospital_share.
During intake	Session cookie is HttpOnly and SameSite=Strict; Secure is used when TLS is configured.
Audio	Temporary recording file is deleted after transcription on success or failure.
Manual clear	Current visit can be cleared; optional registry-erasure path exists for saved details.
Idle	Warning and automatic patient-session clearing protect an unattended kiosk.
After export	The patient kiosk credential is invalidated immediately.
Retention	Configurable registry retention/purge exists; approved clinical/audit records remain according to policy.

Privacy controls

- Local LLM, speech and OCR by default.
- Hashed patient/staff tokens and masked identifiers.
- Scoped patient/staff endpoints and role checks.
- Rate limiting and bounded uploads.
- Explicit correction, sign-off and export audit.

Granular per-scope withdrawal, a patient consent wallet, guardian flows, production encryption at rest and legal certification remain unfinished. Code controls support privacy but do not constitute a compliance certificate.

16 ABHA, ABDM, FHIR and HIS

Layer	Question	MediKiosk today
Medi ID	How does this hospital app recognize the patient?	Local identifier and returning lookup.
ABHA	What national health identity may the patient use?	Optional capture plus staff verification record; no live OTP/QR.
FHIR R4	How is approved health information represented?	Document Bundle with OPConsultRecord Composition first and absolute references.
ABDM	How can records move nationally under consent?	Local M1/M2/M3 state and credential-gated sandbox boundary.
HIS/EMR	Where does the hospital use/store the record?	Destination concept; no live vendor-specific connector.

FHIR resources

Patient, Encounter, Practitioner, Organization, Composition, Condition, MedicationStatement, AllergyIntolerance, Observation and Flag. A local validator fails closed before staging/transport.

Required for live exchange

- Registered facility/bridge context and issued current credentials/paths.
- Real ABHA verification and consent-manager flows.
- Consent-artifact key handling, encryption/decryption and secure transfer.
- Official current NRCeS validation, sandbox testing and certification.
- Inbound signature/provenance validation and clinician review.
- A real HIS/EMR adapter and workflow destination.

Local mode must never be demonstrated as proof that data reached ABDM. It proves mapping, validation and workflow preparation.

17 System architecture

React UI :5173 > FastAPI :8080 > Clinical services > SQLite > Signed outputs

Component	Technology	Responsibility
Patient/staff UI	React + Vite	Touch flow, orb/captions, documents, dashboards, diagnostics and fleet view.
API/controller	FastAPI	Authorization, sessions, clinical schedule, validation, safety, persistence and outputs.
Language model	Ollama llama3.1:8b	Question phrasing and bounded extraction.
Speech	faster-whisper + Piper	Local English/Hindi STT and TTS.
Remote speech adapters	Bhashini / AI4Bharat	Configurable ASR/TTS boundary with local fallback; credentials/endpoints pending.
Documents	EasyOCR + Llama	Recognition, classification and constrained entities.
Database	SQLite	Prototype source of truth for visits, staff, alerts, revisions, audit and integration state.
Output	ReportLab + FHIR JSON	Patient/clinician PDF and interoperable document candidate.

Why this architecture

It is easy to run locally, keeps the clinical controller explicit, separates generative language from authority and supports a convincing offline-focused demonstration. It is not yet a high-availability hospital architecture.

18 Data model and trust

Group	Representative fields	Trust/provenance
Identity	patient_id, session_id, language, mode	Application-owned.
Complaint/HPI	chief complaint and SOCRATES	Patient answer plus transcript.
History	medical, surgical, family, personal, systems	Patient-reported, clinician-correctable.
Medication safety	current medicines and allergies	Asked early; clinician review required.
Ayurveda	Vikriti, Dashavidha, Agni, Koshtha, Nidana	Patient vs practitioner source is separated.
Documents	type/date/status/entities/labs	OCR confidence and review state retained.
Safety	red flag, reason, category	Deterministic/model evidence retained.
Workflow	consent, read-back, status, sign-off	Application and named clinician authority.

Trust ledger

- Captured versus missing clinical coverage.
- Exact transcript alongside structured data.
- Red-flag source and evidence.
- Document confidence and handwriting-review status.
- Revision reason, actor and version.
- Practitioner confirmation source and time.
- Signed snapshot and export validity.

Core rule: confidence is not truth. Patient-reported, OCR-extracted, model-inferred and clinician-confirmed information must remain distinguishable.

19 Multi-kiosk operations and failure behavior

Shared pilot model

Every kiosk points to one central API/database and sends a stable kiosk ID. Sessions, alerts, rate limits, audit and staff accounts share one authority.

Device view

Heartbeats expose online/offline state and workload. Device Check verifies browser access to touch, mic, speaker, camera and network.

Failure	Patient/staff experience
Backend unavailable	Full-screen recovery state blocks unsafe progress and retries.
Ollama unavailable	Non-AI areas remain usable; AI banner explains the limitation.
Unexpected UI crash	Error boundary shows Restart MediKiosk instead of a blank page.
Speech provider failure	Configured remote provider can fall back to local Whisper/Piper.
Unreadable document	Marked illegible/needs staff review; physical copy remains authoritative.
Unacknowledged alert	Visible/audible escalation continues until named acknowledgement.

A real fleet still needs a managed database, encryption at rest, backups/recovery, observability, high availability or degraded offline operation, hospital paging and documented physical commissioning.

20 Current API map - patient and shared services

Method	Path	Purpose
GET	/health	Backend, Ollama, speech, ABDM and deployment status.
POST	/sessions/start	Create consent-backed session.
PATCH	/sessions/{id}/preferences	Save language and interaction mode.
GET	/sessions/{id}	Restore authorized patient visit.
DELETE	/sessions/{id}	Clear current visit.
PATCH	/sessions/{id}/department	Save department.
POST	/sessions/{id}/read-back/confirm	Accept spoken summary.
POST	/sessions/{id}/read-back/dispute	Dispute summary and alert staff.
PUT	/sessions/{id}/documents	Save validated document records.
POST	/patients/register	Create Medi ID.
GET	/patients/{medi_id}	Attach returning patient.
DELETE	/patients/{medi_id}/registry	Patient-requested saved-detail erasure.
POST	/chat	Process one controlled clinical turn.
POST	/transcribe	Audio to text.
POST	/speak	Text to WAV audio.
POST	/ocr	Document image to reviewed structured candidate.
POST	/nurse-station/help-request	Patient asks staff for help.

21 Current API map - staff, clinical and integration

Method	Path	Access / purpose
POST	/staff/login	Staff - issue time-limited role token.
POST	/staff/logout	Staff - invalidate token.
GET	/staff/sessions	Doctor/Admin - list recent visits.
GET	/staff/sessions/{id}	Authorized staff - complete record.
PATCH	/staff/sessions/{id}/clinical-data	Doctor/Admin - correct with reason.
GET	/staff/sessions/{id}/revisions	Doctor/Admin - immutable versions.
POST	/staff/sessions/{id}/ayush-confirmation	Doctor/Admin - protected findings.
POST	/staff/sessions/{id}/sign-off	Doctor/Admin - attest and sign.
GET	/staff/sessions/{id}/pdf/{kind}	Doctor/Admin - patient/clinician PDF.
GET	/nurse-station/alerts	Nurse/Doctor/Admin - active alerts.
POST	/nurse-station/alerts/{id}/acknowledge	Nurse/Doctor/Admin - acknowledge.
GET	/staff/audit	Admin only - newest audit events; no frontend page yet.
GET	/staff/fleet/status	Admin only - kiosk fleet/workload.
PATCH	/staff/patients/{medi_id}/abha	Doctor/Admin - verified link metadata.
POST	/abdm/push	Doctor/Admin - validate/stage or configured transport.
GET	/staff/abdm/status	Doctor/Admin - integration readiness.
POST	/staff/abdm/hiu/consent-requests	Doctor/Admin - bounded HIU request.

Interactive documentation: http://localhost:8080/docs. Backend port is 8080, not 8000.

22 Verification and maturity

43	17/17	6/6	PASS
backend regressions	clinical safety cases	contrast checks	production build

Evidence	What it supports	What it does not prove
Backend regressions	Session, security, clinical, OCR, sign-off and integration contracts.	Real clinical accuracy or hospital workload.
Safety dataset	Selected English/Hindi deterministic examples.	Complete triage sensitivity/specificity.
Frontend build/checks	Current bundle compiles and selected visual rules pass.	All devices, disabilities or environmental conditions.
Browser walkthrough	Light/dark, touch, voice and viewport flows worked in the tested environment.	Physical kiosk commissioning.
FHIR local validator	Internal document invariants fail closed.	Official current NRCeS certification or gateway acceptance.

Pilot metrics

- Completion rate and median intake time.
- History completeness and clinician correction rate.
- Red-flag sensitivity, specificity and alert burden.
- Speech semantic error by language/accent/noise.
- OCR entity and unsafe-error rates.
- Doctor review time and consultation-time effect.
- Consent comprehension and accessibility outcomes.

Current label: hackathon-grade functional MVP with a strong engineering foundation, not a production clinical deployment.

23 Problem statement coverage

Area	Status	Boundary
Adaptive voice + touch	Covered	Hands-free guided Speak and full Chat/touch flow.
Indian languages	Partial	English/Hindi; provider adapters exist; broader languages pending.
SOCRATES	Covered	Server-controlled standard history.
Ayurveda	Strong partial	Full Dashavidha model; protected Trividha/Ashtavidha placeholders; full exam UI pending.
Red flags	Prototype	Expanded categories and escalation; clinical validation pending.
Documents	Partial	Printed OCR works; handwriting reviewed; timeline/procedure depth pending.
Abnormal labs	Partial	High/low flags displayed when extracted; no deterministic range engine.
Drug interactions	Missing	Starter name matching only.
Editable summary	Covered	Reasons, revisions, attestation and sign-off.
Bilingual output	Partial	Patient UI/read-back EN/HI; full clinician document translation pending.
Consent/privacy	Partial	Audio/scopes/clear/retention/erasure; per-scope revoke pending.
ABHA/ABDM/FHIR	Partial	Identity link and local boundary; live certified exchange pending.
Aadhaar	Missing	No eKYC flow.
HIS/EMR	Partial	Outputs exist; real connector pending.

Fair verdict: the product demonstrates all four major modules and most of the intended journey. It does not yet meet every literal requirement at clinical-production depth.

24 Known boundaries

Clinical

No diagnosis/prescribing. Rules, Hindi, OCR and summaries need clinician validation. Handwriting stays under review. No drug-interaction engine.

Integration

No live HIS connector, production ABHA OTP/QR, consent cryptography, official FHIR result, HFR/HPR completion or ABDM certification.

Operations

SQLite, no encryption at rest, no automated backup/DR, no HA/offline queue, no external pager/SMS, no physical commissioning.

Experience

English/Hindi only; remote speech credentials absent; turn-based voice; no longitudinal patient timeline; full bilingual clinician output absent.

Claims to use

- AI-assisted structured history and document intake.
- Clinician-controlled draft, revisions and signed outputs.
- Local English/Hindi voice path and touch fallback.
- Prototype safety alerts with transparent evidence.
- FHIR/ABDM-ready integration boundary in local mode.

Claims to avoid

- Production-ready, fully compliant, clinically validated, diagnostic or autonomous.
- All-language support, reliable handwriting, complete drug safety, live ABDM or live HIS.

25 What to build next

Priority	Work	Success condition
P0	Clinical and usability validation	Real OPD evidence, clinician-approved cases and resolved high-risk failures.
P0	Medication/lab safety	Governed interaction source, explainable alerts and deterministic range logic.
P1	Problem-statement gaps	Aadhaar decision, document procedures, bilingual output and per-scope revoke.
P1	Language breadth	Selected regional languages across UI, consent, ASR/TTS, glossary and OCR.
P1	Ayurveda/AYUSH depth	Full practitioner examination workflow and separate profiles for added disciplines.
P2	Hospital pilot	Production database, mandatory TLS, backup/monitoring, external alerts and HIS adapter.
P2	ABDM live	Registered environment, real identity/consent, cryptography, official validation and certification.
P3	Experience and scale	Streaming voice, offline/degraded mode, longitudinal timeline and patient consent view.

Recommended sequence

Validate workflow > Close safety gaps > Pilot integration > ABDM certification > Scale

The next major proof is not another screen. It is evidence that MediKiosk reduces clinician intake burden without lowering history quality or missing urgent cases.

26 Run, use and demonstrate

Development

Double-click start-dev.bat. Backend runs at http://localhost:8080 and frontend at http://localhost:5173.

Kiosk demonstration

Double-click kiosk-mode.bat. Chrome opens full-screen; Edge is the fallback. Exit with Alt+F4.

Recommended demonstration

- Open Device Check and confirm system health.
- Start, play bilingual consent, choose Speak and show the centered orb/captions/automatic listening.
- Create or retrieve a Medi ID, choose an Ayurveda department and answer a short history path.
- Scan one clean printed record and show the mandatory handwriting-review route.
- Trigger a prepared red flag and show Nurse Station evidence and acknowledgement.
- Open Physician View, correct a field with a reason, inspect the revision, confirm Ayurveda findings and sign.
- Generate patient and clinician PDFs; show the local validated FHIR document while stating that live delivery is pending.
- Use GET /staff/audit in API docs to show attributed events; explain that the frontend audit page is still a known gap.

Before a demo

- Run Ollama with llama3.1:8b and warm Whisper, Piper and EasyOCR.
- Confirm /health, microphone, speaker, camera and selected language.
- Change default staff credentials if the environment is public.
- Use prepared document and red-flag examples; keep the limitations slide/section available.

27 Glossary and references

Term	Plain meaning
OPD	Outpatient Department.
SOCRATES	Eight-part symptom-history framework.
LLM	Large language model used for phrasing and extraction.
STT / TTS	Speech-to-text / text-to-speech.
OCR	Text recognition from document images.
RBAC	Role-based access control.
ABHA	National digital-health identity/account.
ABDM	India's consent-based digital-health ecosystem.
FHIR R4	Healthcare resource format and exchange standard.
HIP / HIU	Provider/user roles in ABDM exchange.
HFR / HPR	Facility/professional registries.
Provenance	Source, actor, time and review state of information.
Dashavidha	Ten-factor Ayurveda assessment.
Trividha / Ashtavidha	Three-method and eight-part Ayurveda examinations.

Primary project references

- ROADMAP.md - current implementation and validation status.
- docs/schema.json - authoritative internal clinical structure.
- docs/AYURVEDA_MODE.md - Ayurveda provenance and protected fields.
- docs/CLINICIAN_WORKFLOW.md - corrections, revisions, sign-off and PDFs.
- docs/ABDM_INTEGRATION.md - identity, FHIR and live-integration gates.
- docs/DEPLOYMENT_AND_COMMISSIONING.md - fleet and device procedure.

External references

- ABDM: https://abdm.gov.in/
- NRCeS ABDM FHIR guide: https://nrces.in/ndhm/fhir/r4/
- HL7 FHIR R4 Bundle: https://hl7.org/fhir/R4/bundle.html
- MeitY DPDP materials: https://www.meity.gov.in/

Final mental model

Patient gives consent, history and documents -> MediKiosk structures them -> deterministic safety watches every turn -> clinician reviews, corrects and signs -> the approved record becomes a patient/clinician PDF or FHIR document -> HIS/ABDM is the exchange destination.

End of current product guide